POOVARASAN M

AI Engineer | Machine Learning Engineer | Data Scientist | Python Developer | Generative AI Engineer

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PROFESSIONAL SUMMARY

Detail-oriented AI/ML Engineer and Data Scientist with an M.Sc. in Information Technology (AI/ML & Data Science) and hands-on expertise in Python, Machine Learning, Deep Learning, Data Analysis, and Generative AI. Proficient in building predictive models, end-to-end ML pipelines, and computer vision systems using TensorFlow, PyTorch, Scikit-learn, Pandas, and NumPy. Skilled in Large Language Models (LLMs), Prompt Engineering, LangChain, and Hugging Face; strong foundation in SQL, Data Visualization, Statistical Modelling, and Software Development. Immediately available for on-site roles; open to relocation.

TECHNICAL SKILLS

Programming Languages: Python, SQL, HTML, CSS

ML / Data Libraries: Scikit-learn, TensorFlow, PyTorch, Keras, Pandas, NumPy, Matplotlib, Seaborn, OpenCV

Machine Learning: Predictive Modeling, Feature Engineering, Model Evaluation, Regression, Classification, Clustering, Cross-Validation, Hyperparameter Tuning

Deep Learning: CNN, RNN, YOLO, Transfer Learning, Backpropagation, Object Detection

Generative AI / NLP: Large Language Models (LLMs), Prompt Engineering, LangChain, Hugging Face, Transformers, NLP

Data Analysis & BI: Data Cleaning, Data Visualization, Exploratory Data Analysis (EDA), Statistical Analysis, Excel, Power BI (Basic), Dashboard Development

Tools & MLOps: Git, GitHub, Jupyter Notebook, VS Code, MLflow, Streamlit, Linux, AWS (Basic)

CS Fundamentals: Data Structures & Algorithms (DSA), OOP, DBMS, Operating Systems, Software Development

Soft Skills: Analytical Thinking, Problem Solving, Teamwork, Communication, Adaptability

EDUCATION

M.Sc. Information Technology (AI/ML & Data Science) | Sri Ramakrishna College of Arts & Science, Coimbatore | 2023 – 2025

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Statistics, DSA, OOP, DBMS

B.Sc. Computer Science | CMS College of Science & Commerce | 2019 – 2022

PROJECTS

Crop Recommendation System | Python, Scikit-learn, Random Forest, Pandas, NumPy, Feature Engineering, EDA

- Developed an end-to-end Machine Learning pipeline for agricultural crop prediction using real-world inputs (NPK, temperature, humidity, pH, rainfall); performed Data Cleaning, Feature Engineering, and Exploratory Data Analysis (EDA).
- Implemented and benchmarked four classification algorithms — Decision Tree, SVM, Logistic Regression, and Random Forest — achieving ~99% Predictive Modeling accuracy; evaluated using Precision, Recall, F1-Score, and Confusion Matrix.
- Designed clean, modular, reusable Python code with Git/GitHub version control, improving experiment reproducibility and Software Development best practices.

Cricket Shot Detection — Real-Time Computer Vision | Python, YOLO, PyTorch, OpenCV, CNN, Transfer Learning

- Architected a real-time object detection system using YOLO and Deep Learning for automated cricket shot classification, reducing manual analysis effort significantly.
- Built the full AI pipeline — data collection, annotation, preprocessing, Model Evaluation, hyperparameter tuning, and inference — using PyTorch and OpenCV on 500+ annotated images.
- Optimized detection performance using CNN and Transfer Learning techniques; maintained a scalable, modular GitHub repository following Analytical Thinking and Software Development standards.

CERTIFICATIONS

- AI with Data Science — CISPRO: Python, Machine Learning algorithms, neural networks, Data Analysis, and Statistical Modelling.
- Full Stack Web Development — EDU TANTR: Front-end/back-end development, database handling, and Software Development fundamentals.

ACHIEVEMENTS & ADDITIONAL INFORMATION

- Achieved ~99% accuracy in crop prediction ML model by engineering optimal features and benchmarking multiple classification algorithms.
- Successfully built and deployed a real-time Computer Vision system processing 500+ images with YOLO, PyTorch, and OpenCV.
- Consistently practising Data Structures & Algorithms (DSA) in Python to strengthen Problem Solving and coding interview readiness.
- Self-learning Generative AI, Large Language Models (LLMs), LangChain, and Hugging Face to stay current with AI advancements.
- Available for immediate on-site joining; open to relocation across India.